

A divisibility that flattens a spectrum

Walsh divisibility $\rightarrow$ three-valued spectrum $\rightarrow$ AB, read slowly

Abstract

One modular fact, on the spectral side.

- `kasami_walsh_div`: $2^{(n+1)/2} \mid W(a, b)$ for the Kasami function.

Combined with two sum identities (**Parseval** and the **fourth moment**), the divisibility pins every Walsh value to $\{0, \pm 2^{(n+1)/2}\}$ — the definition of **Almost Bent (AB)**. This is the spectral twin of the char-2 root lemma: a tiny fact that forces a headline.

Setting: the Walsh transform

Let χ be the sign character of the trace on $F = \text{GF}(2^n)$. The Walsh coefficient of f at (a, b) is the Fourier coefficient

$$W(a, b) = \sum_{x \in F} \chi(ax + bf(x)) \in \mathbb{Z}.$$

```
noncomputable def walsh (f : F → F) (a b : F) : ℤ := ∑
  x : F, χ(a * x + b * f x)

noncomputable def IsAB {n : ℕ} (_ : Fintype.card F = 2 ^ n) (f : F → F) : Prop := ∀
  a : F, a ≠ 0 → ∀ b : F,
    walsh f a b ^ 2 = 0 ∨ walsh f a b ^ 2 = (2 ^ (n + 1) : ℤ)
```

AB says the squared spectrum takes only *two* values: 0 and 2^{n+1} . Equivalently $W \in \{0, \pm 2^{(n+1)/2}\}$ — a three-valued spectrum.

1 The tiny fact: divisibility

```
theorem kasami_walsh_div {n : ℕ} (hcard : Fintype.card F = 2 ^ n)
  (k : ℕ) (hk : k ≥ 1) (hcop : Nat.Coprime k n) (hnodd : Odd n) (hn : n ≥ 1)
  (a b : F) :
  (2 : ℤ) ^ ((n + 1) / 2) | walsh (· ^ d k : F → F) a b
```

Why. Substitute $x = y^{2^k+1}$ (a Gold permutation). The Kasami Walsh sum becomes a sum over a *quadratic* form,

$$W(a, b) = \sum_y \chi(a y^{2^k+1} + b y^{2^{3k}+1}),$$

whose third derivative vanishes. A Gauss-sum/McEliece estimate then gives $2^{(n+1)/2} \mid W$.

This single modular fact is what pins the spectrum to a 2-adic lattice. On its own it says nothing about *which* multiples occur — that needs two moment identities.

2 The two moments (short sum identities)

Write the squared spectrum as a histogram. Two sums constrain it.

Parseval (the L^2 identity)

$$\sum_b W(a, b)^2 = |F|^2 = (2^n)^2 \quad (a \neq 0, f \text{ bijective}).$$

```
theorem parseval_perm {n : ℕ} (hcard : Fintype.card F = 2 ^ n)
  (f : F → F) (hf : Function.Bijective f) (a : F) (ha : a ≠ 0) :
  ∑ b : F, walsh f a b ^ 2 = (Fintype.card F : ℤ) ^ 2
```

Fourth moment (the L^4 identity, from APN)

$$\sum_b W(a, b)^4 = 2 |F|^3.$$

Parseval is Plancherel for the additive Fourier transform: orthogonality of characters ($\sum_x \chi(cx) = 0$ for $c \neq 0$) collapses the double sum. The fourth moment is the autocorrelation/APN identity (the differential “two-solutions” shows up as the constant 2).

3 The squeeze: lattice + two moments = flat

By divisibility, write $W(a, b) = 2^{(n+1)/2} k_b$ with $k_b \in \mathbb{Z}$. Substitute into the two moments (and use n odd):

$$\sum_b k_b^2 = 2^{n-1}, \quad \sum_b k_b^4 = 2^{n-1}.$$

Subtract:

$$\sum_b (k_b^4 - k_b^2) = \sum_b k_b^2 (k_b^2 - 1) = 0.$$

Every summand is ≥ 0 , so each is 0: $k_b^2 \in \{0, 1\}$. Hence

$$W(a, b)^2 \in \{0, 2^{n+1}\}.$$

That is AB. ■

```

theorem ab_from_moments {n : ℕ} (hcard : Fintype.card F = 2 ^ n)
  (f : F → F) (hodd : Odd n) (hn : n ≥ 1)
  (hparseval : ∀ a, a ≠ 0 → ∑ b, walsh f a b ^ 2 = (Fintype.card F : ℤ) ^ 2)
  (hfourth : ∀ a, a ≠ 0 → ∑ b, walsh f a b ^ 4 = 2 * (Fintype.card F : ℤ) ^ 3)
  (hdiv : ∀ a b, (2 : ℤ) ^ ((n + 1) / 2) ∣ walsh f a b) :
  IsAB hcard f

```

Integer lattice + two quadratic constraints $\Rightarrow$ the values sit at ± 1 (after scaling). This is the spectral analogue of “a product is zero $\Rightarrow$ a factor is zero”: $k^2(k^2 - 1) = 0$ over a sum of squares.

4 The headline: APN power permutation, n odd $\Rightarrow$ AB

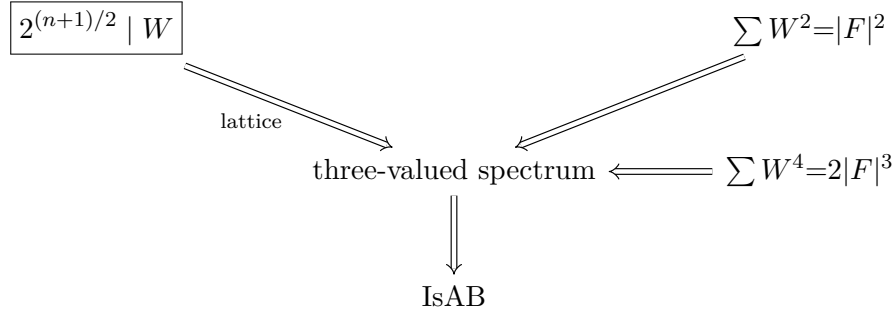
```

theorem apn_implies_ab_power {n : ℕ} (hcard : Fintype.card F = 2 ^ n)
  (d : ℕ) (hd : Nat.Coprime d (Fintype.card F - 1))
  (hbij : Function.Bijective (· ^ d : F → F))
  (hodd : Odd n) (hn : n ≥ 1)
  (hapn : IsAPN (· ^ d : F → F)) :
  IsAB hcard (· ^ d : F → F) := by
  apply ab_from_moments hcard _ hodd hn
  · exact fun a ha => parseval_perm hcard _ hbij a ha
  · exact fun a ha => fourth_moment_apn hcard d hbij hapn a ha
  · exact fun a b => walsh_pow_divisibility hcard d hd hodd hn a b

```

The structural twin of the differential-side “transport” step: the harder *spectral* result (AB) subsumes the easier *differential* one (APN) when n is odd. Three small facts — Parseval, fourth moment, divisibility — assemble into the headline.

The staircase



One-screen summary

fact	statement	role
divisibility	$2^{(n+1)/2} \mid W$	put W on a lattice
Parseval	$\sum_b W^2 = F ^2$	$\sum k_b^2 = 2^{n-1}$
4th moment	$\sum_b W^4 = 2 F ^3$	$\sum k_b^4 = 2^{n-1}$
squeeze	$k_b^2(k_b^2 - 1) = 0$	$W^2 \in \{0, 2^{n+1}\}$ (AB)

$2^{(n+1)/2} \mid W + \text{Parseval} + \text{4th moment} \Rightarrow \text{three-valued spectrum} \Rightarrow \text{AB} \Rightarrow \text{APN}.$